INCH-POUND
MIL-DTL-24643/14G
27 November 2023
SUPERSEDING
MIL-DTL-24643/14F
1 October 2009

DETAIL SPECIFICATION SHEET

CABLE, ELECTRICAL, -4 DEGREES FAHRENHEIT (-20 DEGREES CELSIUS) TO +221 DEGREES FAHRENHEIT (+105 DEGREES CELSIUS), 1,000 VOLTS, TYPE LSSSGU

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24643.

Construction (Watertight with Circuit Integrity)

First - Copper conductor, coated or uncoated (see table I for size).

Second - Thermoset insulation (white), extruded or taped (see table I for thickness).

Third - An optional binder.

Fourth - Cross-linked polyolefin jacket. See table I for thickness.

NOTE: Manufacturer's identification tape is required (see 3.4.9 of MIL-DTL-24643).

AMSC N/A FSC 6145

Distribution Statement A. Approved for public release: distribution is unlimited.

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TABLE I. Details.

Military part no. M24643/14	Type and size	Conductor size		Cable overall diameter max. (inches)	Insulation thickness min. avg. (inch)	Jacket thickness min. avg. (inch)	Conductor resistance max. (ohms)	Insulation resistance (megohms)	Accelerated service loading (amperes)
		Navy STD	AWG/MCM						
-01UN	LSSSGU-50	--	3 (class C)	0.520	0.035	0.209	0.040	160	240
-02UN	LSSSGU-75	--	1 (class C)	0.602	0.035	0.132	0.040	135	315
-03UN	LSSSGU-100	--	0 (class D)	0.669	0.040	0.105	0.050	125	375
-04UN	LSSSGU-200	--	0000 (class D)	0.872	0.050	0.0525	0.050	90	580
-05UN	LSSSGU-300	--	300 MCM (class D)	1.001	0.060	0.0370	0.050	80	750
-06UN	LSSSGU-400	400 (127)	--	1.118	0.060	0.0268	0.050	70	925
-07UN	LSSSGU-650	--	650 MCM (class D)	1.371	0.060	0.0170	0.060	60	1,300
-08UN	LSSSGU-800	--	800 MCM (class D)	1.485	0.060	0.0138	0.060	50	1,550
-09UN	LSSSGU-1000	--	1000 MCM (class D)	1.620	0.060	0.0110	0.060	40	N/A
-10UN	LSSSGU-1600	--	1600 MCM (class B)	2.010	0.080	0.00693	0.075	40	N/A
-11UN	LSSSGU-2000	--	2000 MCM (class B)	2.210	0.080	0.00555	0.075	35	N/A

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REQUIREMENTS:

Qualification required.

INSPECTION:

Basic electricals:

Conductor resistance (ohms/1,000 feet at 77 °F [25 °C], max.)	See table I
Voltage withstand (volts, root mean square, min.)	
Conductor to ground	5,000
Insulation resistance (megohms/1,000 feet, min.)	
Conductor to ground	See table I
Conductor continuity	No failure

Group A:

Visual and dimensional	No failure
Watertightness (see MIL-DTL-24643 for limits of water leakage)	No failure

Group B:

Cold bending (cable)	No failure
Thermoset proof test (percent, max.)	
Insulation (extruded insulation only)	50
Jacket (when tested at 392 °F [200 °C])	50
Drip (203±1.8 °F [95±1 °C])	No failure
Physicals (unaged)	
Insulation (extruded insulation only)	
Tensile strength (lb/in^2, min.)	700
Elongation (percent, min.)	150
Jacket (cable)	
Tensile strength (lb/in^2, min.)	1,300
Elongation (percent, min.)	160
Tear (lb/in thickness, min.)	35

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Group C:

Physicals (aged) (air oven)	
Jacket (cable)	
Tensile strength (percent of unaged, min.)	60
Elongation (percent of unaged, min.)	60
Permanence of printing (jacket) (cycles, min.)	125
Heat distortion (percent of unaged, max.)	30
Cable fill (sealant) removability	No failure
Shrinkage	No failure

Group D:

Gas flame (circuit integrity) (1 hour)	No failure
Flame propagation (cable)	No failure

Qualification inspection:

Qualification inspection shall include basic electricals; groups A, B, C, and D; plus the following:

Cold working (-4±3.6 °F [-20±2 °C])	No damage
Gas flame (circuit integrity) (3 hours)	No failure
Accelerated service (sizes 50 through 800) (see table I for load current)	No failure
Aging and compatibility (cable) (257±9 °F [125±5 °C])	No failure
Abrasion resistance (jacket) (scrapes, min.)	75
Acid gas equivalent (percent, max.)	
Jacket	2
Fillers	2
Insulation	18
Halogen content (percent, max.)	
Jacket	0.2
Fillers	0.2
Insulation	0.2
Immersion (jacket)	
Tensile strength (percent of unaged, min.)	50
Elongation (percent of unaged, min.)	50
Smoke index (max.)	
Jacket	25

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Fillers	45
Insulation	45
Toxicity index (max.)	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) – type A (hardness, min.)	80
Weathering (jacket)	No failure
Electrical moisture absorption (extruded insulation only)	No failure

UNIT ORDERING LENGTHS:

Type	Size	Feet (nominal)
LSSSGU	50 through 100	1,500
LSSSGU	200 through 1,000	1,000
LSSSGU	1,600 and 2,000	500

CHANGES FROM PREVIOUS ISSUE: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Custodians:
Army – MI
Navy – SH

Preparing activity:
Navy – SH
(Project 6145-2013-030)

Review activities:
Army – AR, CR
Navy – CG, EC
DLA – CC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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